

HOMEWORK 3:

OPTIMIZATION AND LINEAR REGRESSION
CPE 486/586

Instructor: Rahul Bhadani

Due: October 11, 2026, 11:59 PM
125 Points

You are allowed to use a generative model-based AI tool for your assignment. However, you must submit an accompanying reflection report detailing how you used the AI tool, the specific query you made, and how it improved your understanding of the subject. You are also required to submit screenshots of your conversation with any large language model (LLM) or equivalent conversational AI, clearly showing the prompts and your login avatar. Some conversational AIs provide a way to share a conversation link, and such a link is desirable for authenticity. Failure to do so may result in actions taken in compliance with the plagiarism policy.

Additionally, you must include your thoughts on how you would approach the assignment if such a tool were not available. Failure to provide a reflection report for every assignment where an AI tool is used may result in a penalty, and subsequent actions will be taken in line with the plagiarism policy.

Submission instruction:

You may either complete your entire homework on Python Notebook and submit a Google Colab link or you may choose a combination of a pdf and Google Colab Notebook. However, you must provide **publicly accessible** Google Colab URL on Canvas.

For .pdf on Canvas, follow the format CPE486586-LastFirst-HW-XX. For example, if your name is Sam Wells, your file name should be CPE486586-WellsSam-HW-XX.pdf.

Your submission must contain your name and UAH Charger ID or your UAH email address. Please number your pages as well.

Please write down your unique keyword corresponding to your package and your Github account URL. Do not provide a link to your Google Colab notebook. Upload your Google colab notebook to your repo in a new folder called HWNotebooks. Provide your GitHub commit hash. Your submitted notebook must be fully executed

with all cells outputs. Please make sure there are no error in your submitted notebook.

Theory

1 Automatic Differentiation and Expression Graph (10 Points)

Consider a function $f(w, x, y, z) = wx + y^2z$. Draw the expression graph corresponding to the given function and demonstrate how would you evaluate $\frac{\partial f}{\partial w}$ for $w = 1$, $x = 2$, $y = 3$, and $z = 4$.

2 Unconstrained Optimization (5 Points)

1. Minimize $f(x) = x^2 - 4x + 4$, determine the critical point, and report the minimum value of the function. **(5 Points)**
2. Optimize $f(x) = x^3 - 3x^2$. Find all the critical points possible. Which one of them are local minima and local maxima? What is the local minimum and maximum values of $f(x)$? **(10 Points)**
3. Consider $f(x, y) = x^2 - y^2$. Compute its gradient ∇f and associated critical point. Is this critical point a maximum, a minimum, or a saddle point? **(5 Points)**

3 Constrained Optimization using Lagrange Multiplier (10 Points)

Using method of Lagrange multiplier, minimize $f(x, y, z) = x^2 + y^2 + z^2$ subject to $x + 2y + z = 6$.

1. Set up the Lagrangian.
2. Derive the system of equations.
3. Solve for critical points.
4. Classify extrema.
5. Explain the role of the constraint.

4 Optimizing a Function using Stochastic Gradient Descent (5 Points)

Use the Stochastic Gradient Descent technique to calculate the optimal value of w_1 for the cost function shown in Equation (1). Use a learning rate $\eta = 0.1$, and an initial value $w_1^{(0)} = 2.0$. Write down the updated value of w_1 for 5 iterations using the formula for gradient descent.

This question is **analytical**, no code is required.

$$J(w_1) = (w_1 - 5)^4 + (w_1 - 3)^2 + 4 \quad (1)$$

How does it compare with optimizing J using partial differentiation method?

5 Linear Regression and Regularization (15 points)

In class we derived and discussed linear regression in detail. Find the result of minimizing the loss of sum of the squared errors after adding in a penalty called an L_2 penalty on the weights. More formally

$$\arg \min_{\mathbf{w}} \left\{ \|\mathbf{y} - \mathbf{xw}\|^2 + \lambda \|\mathbf{w}\|^2 \right\} \quad (2)$$

which means we are interested in finding $\mathbf{w}$ that minimizes the overall expression. How does this change the solution to the original linear regression solution? What is the impact of adding in this penalty?

$\|\cdot\|$ is L2 norm.

Hint: First take the gradient with respect to $\mathbf{w}$ and then equate to zero you will obtain an expression for $\mathbf{w}$ which is the estimated $\hat{\mathbf{w}}$.

Also, remember in vector form the L2 norm $\|v\|$ can be written as $\mathbf{v}^\top \mathbf{v}$.

6 Assumptions in Linear Regression (5 Points)

Write all the assumptions while considering simple linear regression model.

Practice

Note that you don't need to use Google Colab for submission.

7 Computing Derivative using PyTorch Autograd (5 Points)

Consider a following function

$$\tau = 2r \cdot \sqrt{\frac{r}{r_s}} \cdot \sqrt{1 - \frac{r_s}{r}} \quad (3)$$

which is the time experienced by an observer traveling along a specific trajectory in curved spacetime. r is the radial coordinate in a spherical coordinate system and r_s is the Schwarzschild radius.

Write a program in PyTorch to compute the derivative $\frac{d\tau}{dr}$ considering the value of $r = 30000.1$ for $r_s = 0.00887$.

8 Linear Regression in One Variable with Scikit-Learn (35 Points)

Consider the dataset https://github.com/rahulbhadani/Datasets/blob/master/RAV4/2020-04-15-05-12-00_2T3Y1RFV8KC014025_CAN_Messages_5_Hz.csv that contains various driving features from a test driving.

In your Google Colab notebook, use Scikit-learn to create a linear regression model with steer angle as independent variable and yaw rate as response variable. You need to split the dataset into training and test with 25% of the rows set aside as a test data. Report the estimated coefficients of the fitted linear model and R^2 computed using the test data.

9 Linear Regression in One Variable with PyTorch (35 Points)

Consider the dataset <https://github.com/rahulbhadani/Datasets/blob/master/Hydropower.csv>. The main task as a part of this assignment is to create a linear model $y = w_0 + w_1x$ and determine the suitable parameter w_0 and w_1 . For this consider the independent variable x as Benefit-Cost-Ratio (BCR) from the given dataset and the response variable y as AnnualProduction. Using the linear model, we should be able to predict Annual Production given a new BCR.

You need to write LinearRegression class based on gradient descent implemented using PyTorch as a part of your Python package with the following specification:

1. Create a subpackage **model** in your UV Python package. (2 Points)
2. The subpackage should contain a module `regression.py`. (2 Points)

3. Add a `LinearRegression` class to the module with the following requirement:
- (a) `__init__` function, which is a constructor should take following parameters as argument: (i) learning rate, and (ii) number of epochs. Further, `__init__` should initialize w_0 and w_1 , SGD optimizer, and mean-squared error as a loss function, and list or vector to store intermediate w_0 , w_1 , and loss as the training progresses. **(5 Points)**
 - (b) `forward` function should specify the linear model and compute the response y and return it. **(2 Points)**
 - (c) `fit` function should perform machine learning on training dataset provided to the function as an argument. `fit` function also should take test data as an additional argument and after training, it should compute R^2 on the test data and print it. **(5 Points)**
 - (d) `predict` function should predict AnnualProduction given a new BCR data. **(4 Points)**
 - (e) `analysis_plot` function should create several plots on the same figure using subplots such as original data, fitted regression line, and w_0 , w_1 and loss as the training progressed. **(5 Points)**
4. Add appropriate entry to `__init__.py` at subpackage level and at package level to facilitate proper import. **(2 Points)**
5. Build your package after incrementing version number, commit to GitHub, and publish to PYPI. **(3 Points)**
6. Show the complete usage of your `LinearRegression` model with `Hydropower.csv` dataset through a standalone notebook that first installs your package. **(5 Points)**